

General Topology 2

Vincent Wang

April 2025

1 Introduction

Let X be a topological space and $E \subset X$. Then $E \subset \overline{E}$, $E = \overline{E}$ iff E is closed, $\overline{\emptyset} = \emptyset$, $\overline{A \cup B} = \overline{A} \cup \overline{B}$, $\overline{\overline{E}} = \overline{E}$.

Proof: X is closed, so the intersection defining $\overline{E}$ is nonempty. Each set in the intersection contains E , so $E \subset \overline{E}$. Suppose E is closed. Then $\overline{E} \subset E$. Since $E \subset \overline{E}$, we have $E = \overline{E}$. If $E = \overline{E}$, then E is closed since $\overline{E}$ is closed. It follows $\overline{\overline{E}} = \overline{E}$. We prove the last result. $A \subset A \cup B$, so $\overline{A} \subset \overline{A \cup B}$. Similarly, $\overline{B} \subset \overline{A \cup B}$, so $\overline{A} \cup \overline{B} \subset \overline{A \cup B}$. $\overline{A \cup B}$ is closed. Since $A \cup B \subset \overline{A \cup B}$, we have $\overline{A \cup B} \subset \overline{A} \cup \overline{B}$. Hence, $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Let $\bar{\cdot} : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$ be a function such that:

K(a): $E \subset \overline{E}$ for all $E \subset X$.

K(b): $\overline{\overline{E}} = \overline{E}$.

K(c): $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

K(d): $\overline{\emptyset} = \emptyset$.

K(e): E is closed in X iff $\overline{E} = E$.

Then $\bar{\cdot}$ defines a topology on X where $\bar{\cdot}$ is the closure operation.

$\emptyset$ is closed. $X \subset \overline{X} \subset X$, so $\overline{X} = X$ and X is closed. For any finite collection of closed sets A_n , $A_i = \overline{A_i}$ and $A_1 \cup A_2 \cup \dots \cup A_n = \overline{A_1 \cup A_2 \cup \dots \cup A_n} = \overline{A_1} \cup \dots \cup \overline{A_n} = \overline{A_1} \cup \dots \cup \overline{A_n}$. The union is closed. Let A_n be any collection of closed sets. Consider $A = \bigcap A_i$. $A \subset A_i$. Suppose $A \subset B$. Since $B = (B - A) \cup A$, $\overline{A \cup B - A} = \overline{B}$, so $\overline{A} \subset \overline{B}$. $\overline{A} \subset A_i$ for all i , so $\overline{A} \subset A$. However, $A \subset \overline{A}$, so $A = \overline{A}$ and A is closed.

Theorem 3.14: (a). $\text{Fr}(E) \cup E = \overline{E} \cap \overline{X - E} \cup E = (\overline{E} \cup E) \cap (\overline{X - E} \cup E)$. $X - E \subset \overline{X - E}$, so $X = (X - E) \cup E \subset \overline{X - E} \cup E$, so $\overline{X - E} \cup E = X$.

$$\text{Fr}(E) \cup E = \overline{E} \cap X = \overline{E}.$$

$$(b). E - \text{Fr}(E) = E - \overline{E} \cap \overline{X - E} = (E - \overline{E}) \cup (E - \overline{X - E}) = E - \overline{X - E} = E^\circ.$$

$$(c). (E - \text{Fr}(E)) \cup \text{Fr}(E) \cup (X - E - \text{Fr}(X - E)) = E \cup \text{Fr}(E) \cup (X - E - \text{Fr}(X - E)) = X.$$

Suppose in $X = \mathbb{R}^n$ a set A has empty frontier. Then $\overline{A} = A \cup \text{Fr}(A) = A$ and $\overline{X - A} = X - A$. Hence A is open and closed, so $A = \emptyset$ or $A = X$.

$$3A.5: \bigcup U \cup (V \cap A) = \bigcup U \cup \bigcup V \cap A = \bigcup U \cup (A \cap \bigcup V). \bigcup U, \bigcup V \in \tau.$$

$$\begin{aligned} & (U_1 \cup (V_1 \cap A)) \cap (U_2 \cup (V_2 \cap A)) \\ &= (U_1 \cap U_2) \cup (U_1 \cap V_2 \cap A) \cup (U_2 \cap V_1 \cap A) \cup (V_1 \cap V_2 \cap A), \end{aligned}$$

which lies in the extension of τ by A as this element is a union of elements of the extension.

Note τ is a subset of this extension by taking $V = \emptyset$, from which $\emptyset$ and X lie in the extension.

Prove or disprove: (a) $\text{Fr}(\text{Fr}(A)) = \text{Fr}(A)$.

Take $A = \mathbb{Q}$. $\text{Fr}(\mathbb{Q}) = \mathbb{R}$, $\text{Fr}(\mathbb{R}) = \emptyset$.

(b) $\text{Int}(\text{Cl}(A)) = \text{Int}(A)$. False. Take $A = \mathbb{Q}$. $\text{Cl}(\mathbb{Q}) = \mathbb{R}$, $\text{Int}(\mathbb{R}) = \mathbb{R}$. $\text{Int}(\mathbb{Q}) = \emptyset$.

Show: $X - \overline{A} = (X - A)^\circ$.

$$X - \bigcap B = \bigcup (X - B) = (X - A)^\circ.$$

$$X - A^\circ = X - \bigcup B = \bigcap (X - B) = \overline{X - B}.$$

3D: Regularly open and closed sets. An open set G is regularly open if $G = \text{Int}(\text{Cl}(G))$. A closed set F is regularly closed if $F = \text{Cl}(\text{Int}(F))$.

1. The complement of a regularly open set is regularly closed and vice versa.

Suppose G is regularly open, so that $G^{-'-' } = G$, or that $G^{-' -} = G'$. Then $G''^{-' -} = G^{-' -} = G'$, so G' is regularly closed. Conversely, assume F is regularly closed so that $F'^{-' -} = F$. Then $F'^{-' -' } = F'$, so F' is regularly open.

Cantor-Bernstein theorem. Suppose there exist one-to-one functions $f : A \hookrightarrow B$ and $g : B \rightarrow A$. Then there exists bijective $h : A \rightarrow B$.

Let each set A correspond to a set A' such that if $A \subset B$, then $A' \subset B'$. Then there exists a fixed set E . That is, $E = E'$.